

PLYOMETRIC JUMPING PERFORMANCES OF MALE AND FEMALE GYMNASTS FROM DIFFERENT HEIGHTS

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ABSTRACT

Marina, M, Jemni, M, Rodriguez, FA, and Jimenez, A. Plyometric jumping performances of male and female gymnasts from different heights. *JStrength Cond Res* 26(7): 1879–1886, 2012—The objective of this study was to investigate and compare the factors influencing plyometric jumping performance between well-trained gymnasts and a control group. Seventy-six gymnasts and 91 moderately active subjects volunteered to participate in this study. Drop jumps (DJ) were performed from 20-, 40-, 60-, 80-, and 100-cm heights. Flight time (FT) and contact time (CT) were recorded using contact mat. Flight time to contact time (FC) ratio and Bosco expression (BE) were calculated. Male gymnasts scored similar FT to their controls, whereas female gymnasts had significantly longer FT compared with their peers. The gymnasts obtained significantly shorter CT than their control groups, whereas their FC ratios were significantly higher and increased when the height of the drops was close to 60 cm. Moreover, gymnasts' BE was greater in comparison to their respective control groups independent of the drop height. The height of the drop that produced the best FC ratio and BE varied between the groups. The best performances were obtained between 40- and 60-cm drop height for both groups. Female control group showed a trend toward a continuing decline with the increase of the drop height. The best gymnasts (finalists at World Championships) obtained their best performance at 80-cm drop. Flight time is the less discriminating factor distinguishing gymnasts' DJ performances. Considering CT, FC, and BE results all together could better profile the gymnasts rather than taken separately. Bosco expression was shown to be more sensitive to the increase in FT; we suggest BE as the best criteria to assess the appropriate drop height for plyometric training purposes in gymnasts as it has been significantly correlated to FT.

KEY WORDS drop jump, power, stiffness, stretch shortening cycle

INTRODUCTION

Bouncing is one of the most important movements in floor and vault routines and is acquired by gymnasts since a very early age as part of their daily training routines. Because plyometric training jumps take a considerable amount of time within gymnastics, inappropriate planning of these sessions and selection of drop height versus age could have serious implications on the improvement of the jumping performance, health, and safety of the gymnasts. This is precisely why this study attempts, among other objectives, to propose useful criteria and recommendations for choosing the more appropriate drop height to improve plyometric performance. To do so, we choose a contact mat. Since the arrival of force platforms, the use of the contact mats has been reduced, but the latter are still the easiest way to run field tests. Even though force platforms are the favorite tool in a laboratory setting because of their high level of accuracy, strong correlation has been highlighted between both systems ($r = 0.988$) (16). Variety of jumping tests have come up, as described in the literature, since Sargent came up with his first test (14,22,25). Most of the authors emphasized the importance of analyzing the estimated height of the jump and the mean power in many sports: football (8), handball (17), basketball (30), volleyball (6,15), rugby (14), and even weightlifting (7). We argue that only one variable could mislead the investigator, as the real jumping performance of an athlete could be the resultant of multiple factors.

Different variables describing jumping performances have also been suggested by different authors, for example, “motor performance index” (23) and “jump height as an index of muscle power” (24,30); whereas Bosco et al. (6) proposed the “estimated average mechanical power” from the relationship between the flight time (FT) and contact time (CT). Nevertheless, recent studies warn about the unsuitable and incorrect use of the term power, as the generic neuromuscular or athletic performance characteristic, and not the true mechanical definition (20).

Other components, such as the body's vertical velocity at the takeoff and the peak jumping velocity are also very important to consider in studying jumping ability (4). These last 2 variables depend on a single, commonly agreed index: net vertical impulse, which exactly determines vertical jump height (20). Moreover, neuromuscular activation, segmental

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TABLE 1. Physical characteristics of the subjects.*

		<i>n</i>	Age	Body mass (kg)	Height (cm)
Gymnasts					
MG	Male	41	18 ± 4.29	55 ± 12.48	161 ± 11.85
FG	Female	35	11 ± 1.86	30 ± 5.63	135 ± 8.00
Control group					
MC	Male	45	19 ± 3.03	69 ± 7.23	176 ± 6.47
FC	Female	46	13 ± 1.75	46 ± 10.22	155 ± 6.74

*MG = male gymnast; FG = female gymnast; MC = male control; FC = female control.

coordination, and the application of a proper technique are very important to maximize impulse and, for instance, the jumping performance (3,4). This study is field based and will focus only on the parameters directly collected with the contact mat: FT and CT. These 2 factors are adequate to assess many exercises performance (34), and the use of contact mat has been proven as an effective assessment tool for gymnasts, allowing direct and immediate feedback in the gymnasium without altering the daily training (26).

The previous paragraphs attempt to point out that the jumping performance can be assessed through many variables: FT, CT, estimated height and power, and so on.

As the present study attempts to propose useful criteria for choosing the more appropriate drop height during plyometric training in gymnasts, we believe it is of capital importance to know which of these most used variables could be considered as more appropriate criteria. To do this, we should take into account more than one variable. Therefore, the objectives of the study are (a) to establish a plyometric jumping profile for well-trained gymnasts, based on different

relations of FT and CT, that could be useful as a reference for further comparative studies; (b) to compare this plyometric jumping profile between a group of competitive male and female gymnasts and matching control groups of similar age, and (c) to recommend the best criteria for choosing the more appropriate drop height for plyometric training in gymnastics.

Our hypothesis stipulates that gymnasts would perform higher jumping performance

in comparison to the physically active control group. This higher performance should be the resultant of longer FT combined with shorter TC and as a consequence higher values of flight time to contact time (FC) ratio and estimated power (Bosco expression, BE).

METHODS

Experimental Approach to the Problem

To confirm the above hypothesis, we compared male and female gymnasts against controls of similar age. It was expected that gymnasts of both sex could take good advantage of higher drop heights in comparison to their control group. To know which one of the variables was the most objective for deciding an appropriate drop, the drop height was considered as an independent variable. The changes in FT, CT, FC ratio, and BE in the 4 groups were considered the dependent variables. Finally, to check the previous hypothesis, we considered that the first approach to the data, and before doing any comparison, was studying the reliability and repeatability of the tests. The young age of the subjects, the type of training, and important amount of coordination required in the plyometric jumps could influence, in some ways, these 2 factors.

Subjects

One hundred sixty-seven subjects volunteered to participate in this study. The subjects included well-trained male and female gymnasts ($n = 76$) and moderately active group of male subjects and female subjects of similar ages ($n = 91$). The gymnasts train a minimum of 20 h·wk⁻¹ with an average of 6 sessions. All members of this group had a minimum of 4 years of specific gymnastic training history. The control group included actively

TABLE 2. Reliability and repeatability statistics of the drop jumps (from 20- to 100-cm height).*

Group		<i>r</i>	ICC	CV (%)	% Difference between trials, T-test (<i>p</i>)	Effect size (confidence interval)
Male gymnast	Min	0.93	0.96	2.9	2.7–3.6	0.32
	Max	0.98	0.99	4.7	(0.04–0.03)	(0.16–0.49)
Female gymnast	Min	0.89	0.95	2.1	2.7–3.4	0.31
	Max	0.99	0.98	4.7	(0.04–0.02)	(0.20–0.42)
Male control	Min	0.84	0.95	6.8	1.8–4.7	0.23
	Max	0.98	0.97	22.4	(0.04–0.03)	(0.05–0.42)
Female control	Min	0.69	0.92	9.6	1.7–5.8	0.23
	Max	0.97	0.97	22.1	(0.03–0.02)	(0.04–0.41)

*Minimal (Min) and maximal (Max) values observed. ICC = intraclass coefficient correlation; *r* = bilateral Pearson correlation; CV = coefficient of variation.

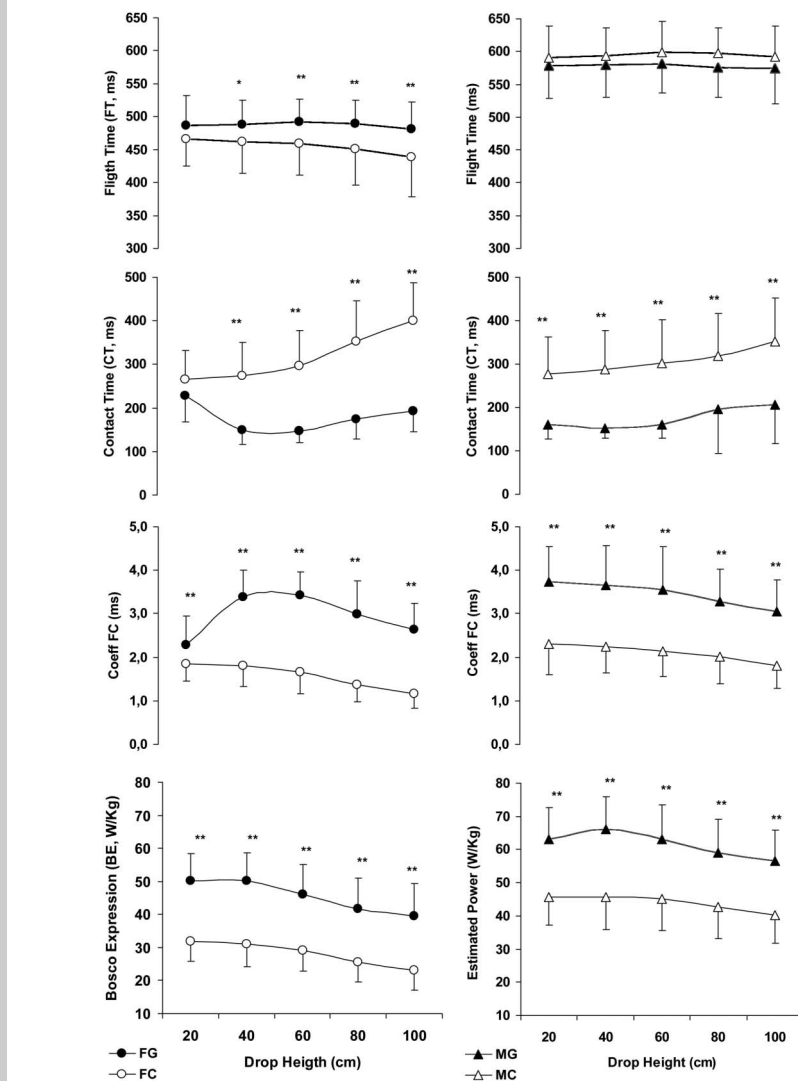


Figure 1. Flight time (FT), contact time (CT), FC, and estimated power (W/kg) during drop jumps at 20-, 40-, 60-, 80-, and 100-cm height. FG = female gymnasts; FC = female control; MG = male gymnasts; MC = male control. * $P \leq 0.01$; ** $P \leq 0.001$.

engaged athletes in football, basket, volleyball, handball, and orienteering. They exercised between 3 and 6 h·wk⁻¹. The tests were carried out during springtime, before the first competitive season and during their usual training time. All subjects were requested to be fully hydrated and ensure that they had their last meal 2 hours before testing. Both groups' descriptive characteristics are in Table 1. The study was approved by the ethics committee of clinical research of Catalan Sport Administration, and written consent were given by all participants and guardians.

Testing Procedure

Subjects were assessed at the gym during the normal training time (late afternoon), without sport shoes, using a contact

mat (1.20 × 0.80 m) connected to an ErgoJump Bosco/System unit. Costumed stairway steps of 20-, 40-, 60-, 80-, and 100-cm height have been used allowing the subjects to jump off (drop jump [DJ]). After 2 or 3 practice trials, subjects performed 2 DJs at each height. One- to 2-minute rest was given between the trials. The DJ that scored the highest mechanical power was considered as the best result and saved for further analysis. To minimize "order effects," the height of the drop was randomly assigned to each subject (a total of 10 jumps per subject were performed in 30 minutes). The contact mat was placed on a standard wood floor and protected by an extra layer of thin carpet to enhance adherence and to reduce the landing impact. Subjects were allowed to help taking off with arms' swing, following the recommendations of Faria and Faria (13). The DJ performed in this study is considered as a modified version of the DJ test published by Komi and Bosco (21). The subjects were instructed to drop as vertically as possible on their toes followed by their heels while minimizing the forward displacement. Then a full extension followed and no leg flexion was allowed during the aerial phase, particularly before

the landing, to minimize the FT biases. In addition, the subjects (mainly the control group) were encouraged to take off immediately after the landing. As a consequence, only a small change in the flexo-extension of the knee angle was allowed (no more than 120 degree) according to Bosco and Komi (5). The main instruction was "perform the maximum flight time (FT) combined with the shortest contact time (CT)" to perform a "quick jump" similar to the ones described by previous studies (3,12,37). The same investigator, who had extensive experience in this type of assessments, ran all tests. If the CT was longer than 400 milliseconds (ms), the trial was then cancelled and the subject was given another chance to repeat the jump (37). This was also the case when incorrect technique was used.

TABLE 3.

Variable in drop jumps	Drop height (cm)	Comparison between male gymnast and control				Comparison between female gymnast and control			
		<i>p</i>	Mean difference (ms)	SE difference (ms)	Confidence interval (ms)	<i>p</i>	Mean difference (ms)	SE difference (ms)	Confidence interval (ms)
Flight time (ms)	20	0.564	5.95	10.26	−14.46 to 26.364	0.047	−19.85	9.84	−39.44 to 0.25
	40	0.325	9.90	9.81	−9.60 to 29.42	0.009	−26.16	9.84	−45.74 to −6.58
	60	0.123	14.97	9.62	−4.16 to 34.01	0.001	−32.72	9.72	−52.07 to −13.37
	80	0.093	15.46	9.09	−2.62 to 33.54	0.001	−38.68	−10.65	−59.89 to −17.47
	100	0.227	12.91	10.60	−8.17 to 33.98	0.001	−43.05	11.94	−66.80 to −19.29
Contact time (ms)	20	0.001	116.55	14.21	88.29 to 144.82	0.013	36.54	14.41	7.87 to 65.23
	40	0.001	134.48	14.86	104.92 to 164.03	0.001	123.18	14.11	95.09 to 151.26
	60	0.001	142.97	16.69	109.78 to 176.16	0.001	148.45	14.29	120.02 to 176.89
	80	0.001	122.35	21.54	79.51 to 165.19	0.001	178.58	17.37	144.01 to 213.15
	100	0.001	146.27	20.38	105.74 to 186.80	0.001	206.89	16.60	173.86 to 239.91
Bosco expression (W/kg)	20	0.001	−18.28	1.90	−22.04 to −14.52	0.001	−18.28	1.59	−21.44 to −15.12
	40	0.001	−21.10	2.11	−25.31 to −16.89	0.001	−19.10	1.72	−22.54 to −15.66
	60	0.001	−18.62	2.12	−22.85 to −14.39	0.001	−16.88	1.74	−20.35 to −13.42
	80	0.001	−16.85	2.08	−20.10 to −12.71	0.001	−16.30	1.73	−19.73 to −12.86
	100	0.001	−16.85	1.93	−20.70 to −13.01	0.001	−16.41	1.80	−19.98 to −12.83
Flight time to contact time ratio (ms)	20	0.001	−1.46	0.16	−1.78 to −1.14	0.001	−0.45	0.12	−0.69 to −0.21
	40	0.001	−1.62	0.12	−1.86 to −1.38	0.001	−1.57	0.13	−1.82 to −1.32
	60	0.001	−1.58	0.14	−1.86 to −1.30	0.001	−1.76	0.12	−1.99 to −1.52
	80	0.001	−1.25	0.15	−1.54 to −0.96	0.001	−1.63	0.13	−1.89 to −1.37
	100	0.001	−1.26	0.14	−1.53 to −0.99	0.001	−1.48	0.11	−1.69 to −1.26

The FT (in milliseconds) and CT (in milliseconds) were automatically measured using the Bosco/System unit.

Mechanical estimation of power was calculated using FT and CT according to the formulae proposed by Bosco et al. (6) (equation 1). This power estimation is presented as BE in this article. We should mention that his equation carries certain known biases as shown by Hatze (19). However, this expression is used in a very strict condition as defended by Arampatzis et al. (2).

$$BE = \frac{g^2 \times FT \times Tt}{4 \times CT} \quad 1$$

Equation 1: Estimated average mechanical power of 1 DJ according to Bosco et al. (6) (Tt = total time of the jump [Tt = FT + CT]; g = gravity force).

Flight time to contact time ratio was calculated; this ratio gives an idea about the direct relation between CT and FT. A high value of FT divided by a reduced CT will result in a high ratio and must be interpreted as a good performance.

Statistical Analyses

The Kolmogorov–Smirnov test was used to assess the normal distribution of the data. A T-test for independent samples was conducted to compare the differences between the gymnasts and the control group. A Levene test was used to check the equality of the variance. One-factor analysis of variance (drop height) for repeated measures followed by Sidack post hoc test for cell to cell comparisons was applied to analyze the performance at each drop's height within a group. Pearson correlation coefficient was also calculated to check the relation between the variables. To check

“test-retest reliability”, an intraclass correlation coefficient (ICC), a coefficient of variation (CV), and a paired-sample T-test between trials were applied. The level of significance was set at 0.05. Statistical analysis was performed with PASW software.

RESULTS

The statistics used to assess the within-subject reliability (Pearson correlation, ICC and paired T-test) confirmed that DJ tests were very reliable in particular within the male and female gymnast groups (Table 2). Otherwise, greater between-subjects variability, assessed by the CV was observed in the control group (6.8–22.4%) in comparison to the gymnasts (2.14–4.69%) (Table 2).

Mean and SD differences between gymnasts and their control group are presented in Figure 1. Descriptive statistical analysis of the mean differences between the groups is presented in Table 3.

Male gymnasts scored similar FT to their control group, whereas female gymnasts had significantly longer FT compared with their peers (Figure 1 and Table 3). In addition, the FT mean difference between female gymnasts and their control group increased with the increase of drop height. Although CT was expected to be reduced in all subjects (considering the above method), the gymnasts of both genders obtained significantly shorter CT than their control groups (Figure 1 and Table 3). A consistent increment of female control group's CT in comparison to the female gymnasts has been noticed. The female control group did not succeed to maintain their CT while the height of the drop was increasing ($F_{2,67,180} = 76.4$; $p < 0.001$) (Figure 1). Furthermore, the following results have been

noted as a consequence of the increase in the CT with the drop height: (a) FC ratios were higher within the gymnast group (both genders) in comparison to the control group and increased when the height of the drops was close to 60 cm ($F_{2,97,292} = 40.68$; $p < 0.001$); (b) BE of both gymnast genders was greater in comparison to their respective control groups independent of the drop height (Figure 1 and Tables 3 and 4). The female gymnasts obtained their best BE at the drop of 40 cm ($F_{3,2,136} = 35.58$; $p < 0.001$), whereas the male gymnasts reached their best BE at the drop of 40 or 60 cm ($F_{4,160} = 17.39$; $p < 0.001$). The control group of both genders achieved their best BE at the lowest

TABLE 4. Bilateral Pearson correlations (*r*) between the drop jump data for 167 subjects. The jumping parameters were correlated for the same drop height independently.*

		FT (ms)	CT (ms)	BE (W/kg)	FC
CT	<i>r</i>	0.02 to 0.13			
	<i>p</i>	0.094 to 0.788			
BE	<i>r</i>	0.71 to 0.82	0.55 to 0.62		
	<i>p</i>	0.001	0.001		
FC	<i>r</i>	0.28 to 0.44	0.80 to 0.89	0.82 to 0.89	
	<i>p</i>	0.001	0.001	0.001	
Body mass (kg)	<i>r</i>	0.55 to 0.59	0.13 to 0.36	0.10 to 0.22	0.09 to 0.21
	<i>p</i>	0.001	0.045 to 0.001	0.162 to 0.007	0.243 to 0.057
Height (cm)	<i>r</i>	0.56 to 0.93	0.16 to 0.40	0.19 to 0.46	0.13 to 0.27
	<i>p</i>	0.001	0.037 to 0.001	0.557 to 0.014	0.049 to 0.001

*FT = flight time; CT = contact time; BE = Bosco expression; FC = flight time to contact time ratio.

drops of 20–40 cm ($F_{2,9,360} = 77.55$; $p < 0.001$).

Correlations between the different data were performed to estimate DJ performance for all subjects (Table 4). We must highlight that CT has a superior correlation with FC ratio in comparison to FT. However, BE was strongly correlated to FT in comparison to CT. Also, it has been noted that body mass and height were poorly correlated or not significantly correlated to CT, BE, and FC ratio, except with a moderate to strong correlation to FT.

DISCUSSION

Because of the great association between jumping performance and segmental coordination, skill and technique (3,4,20,37) and training specificity are important factors to take into account (18), especially when DJ is performed (2). Therefore, it must be expected that gymnasts who practice more than 20 h·wk⁻¹ should have a great influence on the DJ reliability and rebound jumping performance. Although authors have confirmed that gymnasts' performance in countermovement jump is very reliable (31), DJs have been less investigated within this population in comparison to other athletes. The reliability tests did not show any "learning effect" during the assessment. Probably, the big volume of rebound jumps during training (more than 1000 per week) from very young ages may have a decisive influence on plyometric reliability within the gymnasts' group. Thus, one of the findings of the present study is the high reliability of the DJs performed at different heights by the male and female gymnasts. The high CV noticed in the control group may reflect a more pronounced variability compared with the gymnasts' group.

Having discussed the reliability of the DJ and to better understand the results of the present study, we want to point out in the first place some basic considerations about the relationship between FT and CT. Equal FT could be obtained with very different CTs during the DJs. Long CTs reduce considerably BE and FC ratio. In this study, subjects were instructed to jump with a CT as short as possible; therefore, a small change in the flexo-extension knee angle combined with a small knee angular displacement were expected (5). In this condition, subjects must develop an important impulse during the ascendant work phase to reach a reasonable vertical release velocity at takeoff and as a result score a reasonable FT. In other words, to jump high with short CT (short duration of force's application), it is necessary to rapidly increase the force to maximize the net final impulse (20).

Correlation analysis also helps in understanding the FT versus CT relationship and its repercussion on FC ratio and BE. It showed significant relation between FC ratio and CT, whereas no significant relation was noticed between FC ratio and FT. Moreover, there was a significant correlation between FT and BE and a poor correlation between BE and CT. These results suggest that FC ratio is particularly sensitive to short CT than a long FT; whereas,

BE is more sensitive to FT increments. To better understand this relation, we propose an example where CT is constant (200 ms) and FT increases from 500 to 700 ms. Having in mind the BE equation (equation 1), this would give an increase of 80% in BE, whereas FC ratio would only increase by 40%. The overall outcome of this analysis shows that both FC ratio and BE are complementary factors in profiling the jumping performances using a contact mat.

As a consequence of this FT versus CT combination (moderate FT and very short CT), BE and FC ratio were significantly higher in gymnasts. Therefore, to establish a plyometric jumping profile for well-trained gymnasts, it is of primary importance to consider the 4 factors together (FT, CT, FC ratio, and BE).

The results of our gymnasts' FTs are similar to those reported by Faria and Faria, who investigated a similar population (13). However, Young et al. (37) have shown higher CT values than ours (higher than 400 ms vs. less than 200 ms, respectively). We have yet to emphasize the fact that these authors (35) have used a DJ technique that has allowed a longer CT. This technique has been called differently: "DJ for rebound height" (DJ-H) by Young et al. (36), "countermovement drop jump" (CDJ) by Bobbert et al. (3), and "deep drop jump" as mentioned by Eloranta (12). Analysis of the results of this present study may suggest that the control group has adopted a "CDJ" technique seeing the long CT (Figure 1). The short CT of the gymnasts reflects a DJ technique, called "bounce drop jump (BDJ)" (3), "DJ for height-time (DJ-H/t)" (36), or "quick drop jump (QDJ)" (12). Similarly, all these authors have also found long CT values comparable with ours (208–220 vs. 184–195 ms, respectively). The elastic and contractile elements may not only shorten faster in BDJ than in CDJ but also develop larger power output (3). More importantly, a change of strategy from BDJ to CDJ suggests a different neuromuscular pattern during the eccentric phase of the landing just before the takeoff as explained by Dyhre-Poulsen et al. (11). A sharp increase of the CT from a certain drop height may suggest a shift of the jumping strategy from a QDJ to a CDJ.

Although all subjects were instructed to apply a quick takeoff after the drop, the control group failed to perform it. Female control group have almost doubled their CT when compared with female gymnasts (Figure 1). This was similar to a finding reported by Bobbert et al. (3), who observed an increase of the flexo-extension's amplitude in those subjects who did not master the BDJ jumps. This is precisely one of the key points of this article; our results suggest that to compare plyometric performance between groups, it is of utmost importance to have a global vision taking into account the 4 parameters, such as, CT, FC ratio, and BE. Thus, FT must not be used in an isolated way in such comparison.

Some authors found a strong correlation between squat jump and CDJ ($r = 0.87$) (37). They suggested that when

a subject switches to a CDJ technique from a certain drop height, he/she would not continue to rely on the reactive component of the stretch-shortening cycle, but he/she would mainly rely on the force-velocity component (3). Schmidbleicher (28) reported that if CT of a subject is longer than 250 ms, the maximal force has a stronger influence on the jumping performance.

Because of the temporal restrictions on jumping performance in gymnastics (1), a gymnast has to learn how to apply a very high level of force in a very short time. This is the reason why some studies related the jumping performance to the rate of force development, especially in the first 40 ms (10,28). According to Bencke et al. (2), it takes about 750 ms to reach maximum isometric force, with male subjects reaching their 70% in 376 ± 256 ms and female subjects in 748 ± 344 ms. If we apply these figures to gymnastics, we would conclude that these time constraints are too short for gymnasts to express their maximum force because tumbling takeoffs are less than 150 ms. Therefore gymnasts, more than other athletes, need to reach a high level of their rate of force development in an extremely short time when performing their daily routines. Without a doubt, neuromuscular facilitation plays a key role here as several authors have demonstrated. Schmidbleicher et al. (29), for example, found that the electromyogram (EMG) recorded in untrained subjects during the descendent phase of a DJ (eccentric) had an inhibition period; meanwhile, trained jumpers' EMG showed a neuromuscular activation phase.

Our study shows that gymnasts are characterized by a high FC ratio in DJ. This was made possible because of the decreased eccentric phase of the jump and reflects a great reactivity and stiffness (35). Our results and discussion point out that FT is the less discriminating factor among the variables examined in the current study when it comes to distinguishing gymnasts' DJ performances to the control group and possibly other sports. This factor should carefully be taken into consideration when comparing similar "highly experienced athletes in jumping."

The results of the present study indicate that the height of the drop that produced the highest FC ratio and BE was not the same between the groups. This could be the result of a combination of multifactors, among them maximal force (9,28,33), stiffness (33,36), neuromuscular preactivation (12,29), potentiation of elastic and reactive components (5,33), and technique (3,4,20,37). All these factors could be improved by training; this in fact may explain the better results of the gymnasts compared with the controls. Taking into consideration all the plyometric data, female control group's performance showed a trend toward a continuing decline with the increase of the drop height. The rest of the groups reached their best performance between 40- and 60-cm drop. It is noteworthy that only the best elite gymnasts (finalists of the World and European Championships) obtained their best BE, FT, and FC ratio at 80-cm drop.

An example was BE = 104 W/kg; FT = 616 ms; CT = 105 ms; FC ratio = 5.9.

Previous studies found considerable intersubject variability of the best drop height that has produced the longest FT (25). Viitasalo et al. (32) consider the optimum drop height as the one that produces the longest FT in any kind of jump (CDJ or QDJ). Meanwhile, Young et al. (36) disagreed and suggested that the longest FT should only be considered for CDJ. The same authors recommended the use of the higher FC ratio during QDJ to choose the suitable drop height, because of the FC ratio's relation to the great demand of muscle stiffness during a BDJ (36).

A significant correlation was noticed between FT and BE. Bosco expression was shown to be more sensitive to FT increments rather than the CT. As we mentioned above, an FT increment from 500 to 700 ms would provoke an increase of 80% in BE. For this reason, instead of FT, we suggest the maximal BE as the best criteria to choose the suitable drop height for plyometric training purposes. Without a significantly long FT, the gymnasts do not have enough time to perform aerial acrobatic figures. This choice can be considered as a compromised position between the criteria outlined above.

PRACTICAL APPLICATIONS

The coaches should not take the FT as the "gold" variable to evaluate the jumping performance of their athletes. Scoring a good FT (correlated to jump height) did not distinguish well-trained competitive gymnasts compared with control groups, particularly the male gymnasts. On the contrary, both gender gymnasts used very short CT to perform their DJs and thus allowed them to obtain very good scores in FC ratio and estimated power (BE). Although the best elite gymnasts of both genders obtained their best performance at 80-cm drop, we suggest the drop heights of 40 and 60 cm as the more appropriate for the majority of the gymnasts. Finally, 3 practical recommendations can be drawn for training purposes: (a) Excessive drop heights should be avoided to preserve plyometric technique as this may increase the CT and therefore the jump would not be considered as "Quick Drop Jump." (b) We recommend the BE as the main criteria to assess individually the most appropriate drop height for plyometric training purposes with gymnasts. In other words, you can increase the drop height as long as the subject can improve his/her BE. However, as soon as the BE declines, you should select the previous drop height for plyometric training. (c) Instead of FT alone, FC ratio and BE could provide more accurate perspectives to compare plyometric jumping performance between subjects.

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